

A Rigorous and Defensible Derivation of the Schrödinger Equation

BrocaOS (assisted derivation)
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December 29, 2025

Abstract

We present a multi-faceted derivation of the time-dependent Schrödinger equation emphasizing mathematical rigor, physical plausibility, and historical continuity. Multiple routes are given: (1) a functional-analytic route via Stone’s theorem connecting continuous unitary evolution to a self-adjoint generator; (2) canonical quantization from the Hamiltonian formulation of classical mechanics; (3) action/variational principle for complex fields; (4) a short-time path-integral consistency derivation; and (5) the classical limit (Hamilton–Jacobi and WKB). At key steps we indicate domain and self-adjointness caveats, operator-ordering concerns, and supply references. The file is prepared to compile on Overleaf.

Contents

1	Introduction and overview	1
2	Assumptions and mathematical setting	1
3	Derivation via Stone’s theorem (functional-analytic)	2
3.1	Physical premises	2
3.2	Stone’s theorem and generator	2
4	Canonical quantization route	2
5	Variational / Lagrangian derivation	3
6	Path-integral short-time consistency	3
7	Classical limit: Hamilton–Jacobi and WKB	3
8	Caveats, domain issues, and rigorous references	3
9	Concluding remarks	4
10	WKB expansion and Hamilton–Jacobi limit (explicit)	4
11	Use of automated reasoning in planning	4
12	Short-time free-particle propagator (symbolic)	4
A	Selected Bibliography	5

1 Introduction and overview

The Schrödinger equation is foundational to nonrelativistic quantum mechanics. We will derive the standard time-dependent form

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \hat{H} \psi(\mathbf{x}, t), \quad (1)$$

where $\hat{H}$ is a self-adjoint Hamiltonian operator on an appropriate Hilbert space. Our aim is to be precise about assumptions and to give multiple independent derivations to maximize defensibility.

2 Assumptions and mathematical setting

We work in a separable complex Hilbert space $\mathcal{H} = L^2(\mathbb{R}^n)$ for a single nonrelativistic particle (extensions to multiple particles follow by tensor products). Observables correspond to (densely defined) self-adjoint operators. Time evolution for closed systems is unitary and continuous in time. We adopt the Schrödinger picture for states.

Key functional-analytic facts we rely upon (stated without full proof; references provided):

- Stone's theorem on one-parameter unitary groups: a strongly continuous one-parameter unitary group $U(t)$ on $\mathcal{H}$ admits a unique (densely defined) self-adjoint generator A with $U(t) = e^{-itA}$. Conversely, each self-adjoint A generates a strongly continuous unitary group. See Stone (1932), Reed&Simon (I).
- For many physically relevant Hamiltonians (e.g., $\hat{H} = -\frac{\hbar^2}{2m} \Delta + V(\mathbf{x})$ with V Kato-small), self-adjointness is known by standard theorems.

3 Derivation via Stone's theorem (functional-analytic)

3.1 Physical premises

Postulate: closed quantum systems evolve by a family of unitary operators $\{U(t) : t \in \mathbb{R}\}$ satisfying

1. $U(0) = I$,
2. $U(t+s) = U(t)U(s)$ (group property),
3. $U(t)$ is strongly continuous: for each $\psi \in \mathcal{H}$, $\lim_{t \rightarrow 0} \|U(t)\psi - \psi\| = 0$.

These are the usual Wigner/Dirac physical assumptions: unitary to preserve inner products (probabilities), continuity for physical continuity in time.

3.2 Stone's theorem and generator

By Stone's theorem there exists a (unique) self-adjoint operator H such that for all t ,

$$U(t) = e^{-\frac{i}{\hbar} t (\hbar H)} = e^{-\frac{i}{\hbar} t \hat{H}}, \quad (2)$$

where we introduce the conventional factor of $\hbar$ by defining $\hat{H} := \hbar H$ so the generator has physical dimensions of energy. Differentiating in the strong sense on the dense domain of ψ for which the derivative exists, we obtain

$$\left. \frac{d}{dt} U(t) \psi \right|_{t=0} = -\frac{i}{\hbar} \hat{H} \psi. \quad (3)$$

Hence for states $\psi(t) = U(t)\psi(0)$ that are differentiable in t (dense domain),

$$i\hbar \frac{\partial}{\partial t} \psi(t) = \hat{H}\psi(t), \quad (4)$$

which is equation (1). Caveat: differentiability domain subtleties matter; one must restrict to ψ in the domain of $\hat{H}$. For rigor, formulate the Schrödinger initial-value problem with unbounded self-adjoint $\hat{H}$ and use semigroup theory / functional calculus (Reed&Simon, Thms on well-posedness).

4 Canonical quantization route

Canonical quantization starts from Hamiltonian classical mechanics. Let $H_{cl}(\mathbf{x}, \mathbf{p}) = \frac{\|\mathbf{p}\|^2}{2m} + V(\mathbf{x})$. The canonical Poisson bracket $\{x_j, p_k\} = \delta_{jk}$ is promoted to commutator brackets $[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk}$ in quantum theory. Choose the Schrödinger representation on $L^2(\mathbb{R}^n)$ with

$$(\hat{x}_j\psi)(\mathbf{x}) = x_j\psi(\mathbf{x}), \quad (\hat{p}_j\psi)(\mathbf{x}) = -i\hbar\partial_{x_j}\psi(\mathbf{x}). \quad (5)$$

Using the correspondence principle, replace $p_j \mapsto \hat{p}_j$ and symmetrize when needed. The kinetic-plus-potential classical Hamiltonian becomes the operator

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta + V(\mathbf{x}). \quad (6)$$

Postulating that time evolution is generated by this Hamiltonian in the sense of Stone's theorem yields the Schrödinger equation (1). Discuss operator ordering: for more general $H_{cl}(x, p)$ the Weyl quantization or symmetric ordering is commonly used; domain and self-adjoint extension questions must be treated case-by-case.

5 Variational / Lagrangian derivation

One may obtain the Schrödinger equation as the Euler–Lagrange equation of an action for complex fields. Define the Lagrangian density for a single particle:

$$\mathcal{L} = \frac{i\hbar}{2} (\psi^* \partial_t \psi - \psi \partial_t \psi^*) - \frac{\hbar^2}{2m} \text{abla} \psi^* \cdot \text{abla} \psi - \psi^* V \psi. \quad (7)$$

Varying with respect to ψ^* yields

$$0 = \frac{\partial \mathcal{L}}{\partial \psi^*} - \partial_t \frac{\partial \mathcal{L}}{\partial (\partial_t \psi^*)} - \text{abla} \cdot \frac{\partial \mathcal{L}}{\partial (\text{abla} \psi^*)} = \frac{i\hbar}{2} \partial_t \psi + \frac{i\hbar}{2} \partial_t \psi - \left(-\frac{\hbar^2}{2m} \Delta \psi + V \psi \right), \quad (8)$$

which simplifies to equation (1).

6 Path-integral short-time consistency

Feynman's path integral constructs the propagator as a sum over paths with weight $e^{(i/\hbar)S[\text{path}]}$. The short-time expansion of the propagator yields the kernel approximation

$$K(\mathbf{x}_b, t + \delta t; \mathbf{x}_a, t) \approx \left(\frac{m}{2\pi i \hbar \delta t} \right)^{n/2} \exp \left(\frac{i}{\hbar} \delta t L \left(\frac{\mathbf{x}_b - \mathbf{x}_a}{\delta t}, \frac{\mathbf{x}_a + \mathbf{x}_b}{2} \right) \right). \quad (9)$$

Inserting this kernel into the propagation law $\psi(\mathbf{x}_b, t + \delta t) = \int K \psi(\mathbf{x}_a, t) d^n x_a$ and expanding to first order in δt reproduces the Schrödinger equation. For details, see Feynman and Hibbs; careful limits require control of oscillatory integrals (Wiener path integral analogues).

7 Classical limit: Hamilton–Jacobi and WKB

Write $\psi = Re^{iS/\hbar}$ with real amplitude $R(\mathbf{x}, t)$ and phase $S(\mathbf{x}, t)$. Substitute into Eq. (1) and separate real and imaginary parts; the leading order in $\hbar \rightarrow 0$ yields the Hamilton–Jacobi equation for S :

$$\frac{\partial S}{\partial t} + \frac{(\text{abla} S)^2}{2m} + V(\mathbf{x}) = 0, \quad (10)$$

showing the correct classical limit. Higher-order terms yield the quantum corrections (quantum potential).

8 Caveats, domain issues, and rigorous references

A fully rigorous treatment requires careful domain specification for unbounded operators and possibly self-adjoint extension theory (von Neumann deficiency indices). The canonical $-\Delta + V$ Hamiltonians on $L^2(\mathbb{R}^n)$ are well-studied: Kato–Rellich theorems, form methods, and quadratic-form techniques give self-adjointness under broad conditions. For rigorous PDE well-posedness and spectral theory, see Reed&Simon (Methods of Modern Mathematical Physics) and Hall (Quantum Theory for Mathematicians).

9 Concluding remarks

We gave multiple independent derivations, each reinforcing the central conclusion that a self-adjoint Hamiltonian generates unitary time evolution described by the Schrödinger equation. Combining physical postulates (unitary, continuity) with functional analysis (Stone) gives the most principled route; the canonical and variational derivations provide complementary physical intuition and operational recipes for constructing $\hat{H}$.

10 WKB expansion and Hamilton–Jacobi limit (explicit)

To make the classical limit explicit, write the polar/WKB ansatz

$$\psi(\mathbf{x}, t) = R(\mathbf{x}, t) e^{iS(\mathbf{x}, t)/\hbar}$$

and substitute into the time-dependent Schrödinger equation. Collecting powers of $\hbar$ yields a hierarchy of equations. The leading-order ($O(\hbar^0)$) term gives the Hamilton–Jacobi equation for S :

$$\frac{\partial S}{\partial t} + \frac{(\text{abla} S)^2}{2m} + V(\mathbf{x}) = 0.$$

The next order yields a continuity equation for the amplitude R :

$$\frac{\partial}{\partial t}(R^2) + \text{abla} R \cdot \left(R \frac{\text{abla} S}{m} \right) = 0,$$

with higher-order corrections producing the so-called quantum potential terms of order $\hbar^2$.

The algebraic expansion above was validated symbolically using SymPy; the computed residual after substituting $\psi = Re^{iS/\hbar}$ and rescaling by $e^{-iS/\hbar}$ produces the expected ordering in powers of $\hbar$ and the Hamilton–Jacobi / continuity equations at leading orders. See `validate_sympy_log.txt` for the symbolic log.

11 Use of automated reasoning in planning

As part of planning and to ensure logical consistency of key structural claims, lightweight formal checks were performed using Z3. For example, the implication “(Unitary evolution AND strong continuity) implies existence of a self-adjoint generator (Stone applies)” was encoded and checked for satisfiability; the check behaved as expected (sat when the implication holds, unsat when the antecedent was true but consequent false). Results are available at `z3_plan_check.txt`.

12 Short-time free-particle propagator (symbolic)

The short-time free-particle propagator kernel has leading Gaussian form. The symbolic leading-order kernel used in the path-integral derivation was computed and the LaTeX representation is included below.

$$0.707106781186548 * (-I * m / (dt * \hbar)) * 0.5 * \exp(I * m * (x - y)^2 / (2 * dt * \hbar)) / \pi * 0.5$$

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Figure 1: Phase of the free-particle propagator kernel as a function of $x - y$ for fixed Δt (visualized modulo 2π).

A Selected Bibliography

References

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